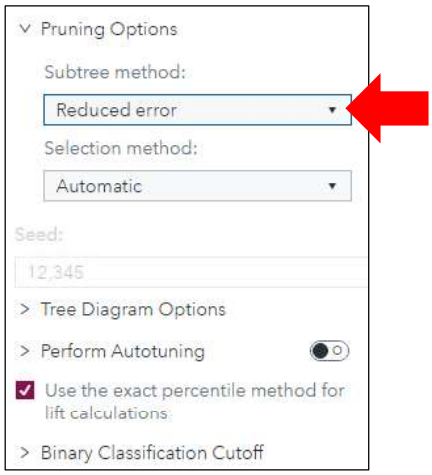
Improving a Decision Tree Model by Changing the Pruning Parameters

In this demonstration, you change the default settings of the Decision Tree node in the Lesson 3 pipeline. You modify the pruning parameters and compare this model performance to the model built earlier in the course.

1. To recall, the previous model, based on changings on the tree structure and the recursive partitioning parameters, achieved an average squared error of 0.0630 on the VALIDATE partition.
2. Try to improve the model's performance by modifying some of the settings of the decision tree model. In the properties pane, expand the properties under Pruning Options.
3. Change **Subtree method** from Cost complexity to **Reduced error**.



The cost complexity pruning method helps prevent overfitting by making a trade-off between the complexity (size) of a tree and the error rate. Thus, large trees with a low error rate are penalized in favor of smaller trees. On the other hand, reduced error subtree method performs pruning and subtree selection based on minimizing the error rate in the validation partition at each pruning step and then in the overall subtree sequence. The error rate is based on the misclassification rate for a categorical response variable.

4. Run the **Decision Tree** node.
5. Open the results for the node.
6. Click the **Assessment** tab. (You might need to expand the Fit Statistics table.)

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0574
churn	VALIDATE	0	0	16,967	0.0626

The average squared error for the tuned decision tree model is 0.0626 on the VALIDATE partition. This average squared error shows the slightest of improvements from the last model built.

6. Close the Results window.

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7. Run the entire pipeline and view the results of the Model Comparison node. The Model Comparison table shows that the decision tree model is currently the champion from the Lesson 3 pipeline. This is based on the default fit statistic KS.

Champion	Name	Algorithm Name	KS (Youden)	Misclassification Rate
★	Decision Tree	Decision Tree	0.5480	0.0711
	Logistic Regression	Logistic Regression	0.5338	0.0815

8. Close the results of the Model Comparison node.

End of Demonstration

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